

Bragg condition and can be used to form an optical cavity in a laser. The method is known as distributed feedback (DFB) or a distributed Bragg reflector (DBR), depending on the configuration. Such lasers have a smaller linewidth than those using a conventional Fabry–Pérot cavity. Superlattices can also be formed by growth, as in Section 3.4.4, and used as distributed Bragg reflectors to form the cavity in vertical surface-emitting lasers.

A layered superlattice generates a band gap only for motion in one direction: there is no effect parallel to the layers. Further patterning to give a three-dimensional periodic potential, as in a solid crystal, may produce a range of frequencies or *photonic band gap* in which no modes of light can propagate. These devices are the subject of active research because of the control they offer over the optical properties, particularly spontaneous emission, which may lead to improved optoelectronic devices.

This concludes our review of the vast range of heterostructures in use. Before moving on to study their electronic structure, we shall take a diversion to explore the effective-mass approximation, a foundation of the theory of inhomogeneous semiconductors.

3.10 Effective-Mass Approximation

One of the simplifications that has been made throughout the foregoing survey is to assume that the band structure of the crystalline semiconductors can largely be ignored, retaining only the energy and effective mass at the extremity of each band. We shall now explore this *effective-mass approximation* to find its limitations and verify where it can be used safely. This section could be omitted on a first reading.

Suppose that we add a perturbation such as an impurity to a perfect crystal. More generally, the perturbation could be a quantum well, barrier, superlattice, or the potential from a patterned gate in a two-dimensional electron gas. The Schrödinger equation takes the form

$$[\hat{H}_{\text{per}} + V_{\text{imp}}(\mathbf{R})]\psi(\mathbf{R}) = E\psi(\mathbf{R}). \quad (3.2)$$

Here $\hat{H}_{\text{per}}$ is the Hamiltonian for the perfect crystal and $V_{\text{imp}}(\mathbf{R})$ is the additional contribution from the impurity. It is difficult to solve the Schrödinger equation for a perfect crystal, so equation (3.2) presents a formidable challenge.

On the other hand, elementary textbooks give simple results for a donor or acceptor in a semiconductor. A donor behaves as though it has an extra positive ionic charge compensated by an extra electron, with the signs reversed for acceptors. The impurity therefore behaves like a hydrogen atom, but the mass is rescaled to the effective mass of electrons or holes, and the charge is screened by the dielectric constant of the host. Both effects reduce the ionization energy substantially from 13.6 eV in hydrogen to only 5 meV for a donor in GaAs. This simple procedure can be justified by effective-mass (or effective-Hamiltonian) theory. It is attractive that

the theory yields so straightforward a result, but it is also important to realize its limitations.

Consider a one-dimensional system to simplify the notation. Suppose that we have solved the Schrödinger equation for the perfect crystal

$$\hat{H}_{\text{per}}\phi_{nk}(x) = \varepsilon_n(k)\phi_{nk}(x). \quad (3.3)$$

These solutions form a complete set (Section 1.6). The wave function $\psi(x)$ of the system with the impurity can therefore be expanded in terms of $\phi_{nk}(x)$ as

$$\psi(x) = \sum_n \int_{-\pi/a}^{\pi/a} \tilde{\chi}_n(k) \phi_{nk}(x) \frac{dk}{2\pi}, \quad (3.4)$$

where $\tilde{\chi}_n(k)$ are expansion coefficients. This has both a summation over all bands n and an integral of k over the Brillouin zone to include all states. We could substitute the expansion (3.4) into the Schrödinger equation (3.2) and attempt to find $\tilde{\chi}_n(k)$. This is, of course, still an exact solution and is no easier to do in principle than solving the original equation directly. The wave function must be simplified to make further progress. We shall make a set of drastic approximations to get a straightforward result, and review afterwards how they may be relaxed.

The first step is to assume that wave functions from only one band play a significant part, so the summation over n can be dropped. Taking a donor in GaAs as an example, we expect the states to be drawn largely from the bottom conduction band (Γ_6). The valence and higher conduction bands should make only a small contribution. This has to be justified a posteriori: we find that the energy of the donor is close to the edge of the conduction band but a long way from the other bands, so the approximation is probably good.

The second step is to assume that states from only a small region of k -space contribute significantly to the integral. In the case of a donor, we expect that k lies around 0 as the bottom of the conduction band is at Γ . Now the Bloch functions can be written as $\phi_{nk}(x) = u_{nk}(x) \exp(ikx)$, where $u_{nk}(x)$ is a periodic function of x . Assume that most of the variation in $\phi_{nk}(x)$ with k comes from the plane wave and that $u_{nk}(x)$ can be treated as independent of k over a small region of k -space. Thus we write

$$\phi_{nk}(x) = u_{nk}(x) e^{ikx} \approx u_{n0}(x) e^{ikx} = \phi_{n0}(x) e^{ikx} \quad (3.5)$$

for small values of k .

With these two simplifications, the wave function (3.4) takes the form of an inverse Fourier transform,

$$\psi(x) \approx \phi_{n0}(x) \int_{-\pi/a}^{\pi/a} \tilde{\chi}(k) \exp(ikx) \frac{dk}{2\pi} = \phi_{n0}(x) \chi(x). \quad (3.6)$$

This is the first major result of effective-mass theory, illustrated in Figure 3.21. The wave function can be written approximately as the product of the Bloch function

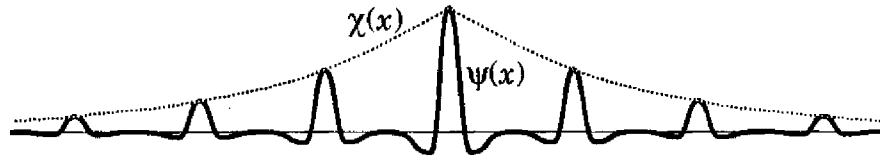


FIGURE 3.21. Wave function around an impurity, showing the envelope function $\chi(x)$ that modulates the Bloch function to give the full wave function $\psi(x)$.

of the local extremum of the host's energy bands and an *envelope function* $\chi(x)$. We assumed that $\tilde{\chi}(k)$ contains only a small range of wave numbers, which in turn means that $\chi(x)$ must be a slowly varying function in real space. Again this must be checked afterwards: the radius of a donor is about 10 nm in GaAs, which is large compared with the lattice constant of 0.5 nm.

This is an attractive step forwards, and we next need an equation for the envelope function. Substitute the expansion (3.4) into the Schrödinger equation (3.2). The main problem is the effect of $\hat{H}_{\text{per}}$. If we make the reduction to a single band, this operator gives

$$\begin{aligned}\hat{H}_{\text{per}}\psi(x) &= \hat{H}_{\text{per}} \int_{-\pi/a}^{\pi/a} \tilde{\chi}(k) \phi_{nk}(x) \frac{dk}{2\pi} = \int_{-\pi/a}^{\pi/a} \tilde{\chi}(k) \varepsilon_n(k) \phi_{nk}(x) \frac{dk}{2\pi} \\ &\approx \phi_{n0}(x) \int_{-\pi/a}^{\pi/a} \tilde{\chi}(k) \varepsilon_n(k) e^{ikx} \frac{dk}{2\pi}.\end{aligned}\quad (3.7)$$

The Hamiltonian $\hat{H}_{\text{per}}$ acts on x and therefore affects only $\phi_{nk}(x)$. We know that this is an eigenfunction of the Schrödinger equation for the pure crystal, equation (3.3), so the operator can be replaced by its eigenvalue $\varepsilon_n(k)$. Finally, we make the approximation (3.5) for the Bloch functions.

Next, expand the energy band as a power series in k : $\varepsilon_n(k) = \sum_m a_m k^m$. This gives

$$\hat{H}_{\text{per}}\psi(x) \approx \phi_{n0}(x) \sum_m a_m \int_{-\pi/a}^{\pi/a} \tilde{\chi}(k) k^m e^{ikx} \frac{dk}{2\pi}.\quad (3.8)$$

Now we use the standard result for the Fourier transform of a derivative, obtained by integration by parts,

$$\int \frac{df(x)}{dx} e^{-ikx} dx = ik \int f(x) e^{-ikx} dx = ik \tilde{f}(k).\quad (3.9)$$

Thus the inverse Fourier transform of $k \tilde{f}(k)$ is $-i df(x)/dx$, and this can be generalized to show that the inverse transform of $k^m \tilde{f}(k)$ is $(-i d/dx)^m f(x)$. Thus equation (3.8) becomes

$$\hat{H}_{\text{per}}\psi(x) \approx \phi_{n0}(x) \sum_m a_m \left(-i \frac{d}{dx}\right)^m \chi(x) \equiv \phi_{n0}(x) \varepsilon_n \left(-i \frac{d}{dx}\right) \chi(x).\quad (3.10)$$

The weird expression $\varepsilon_n(-i d/dx)$ is no more than an abbreviation for the expansion on the left. It means that one should take $\varepsilon_n(k)$, expand it as a power series in k , and replace k with $-i d/dx$ in every term. In general there is an infinite number of terms so $\varepsilon_n(-i d/dx)$ is a nasty expression despite its neat appearance.

We have thus reduced the Schrödinger equation (3.2) to a remarkably simple form because the remaining terms $V_{\text{imp}}\psi$ and $E\psi$ simply multiply the wave function. The common factor of $\phi_{n0}(x)$ cancels to leave

$$\left[\varepsilon_n \left(-i \frac{d}{dx} \right) + V_{\text{imp}}(x) \right] \chi(x) = E \chi(x). \quad (3.11)$$

This is our final result: a Schrödinger equation for the envelope function alone, containing an *effective Hamiltonian*. The Bloch functions have vanished, as has the periodic potential of the host, in favour of a complicated kinetic energy operator that contains the band structure. In three dimensions, $\varepsilon_n(-i d/dx)$ is replaced by $\varepsilon_n(-i \nabla)$. The normalization of the envelope function is a minor problem, because the full wave function is a product of χ and ϕ_{n0} , and will be addressed in Section 10.4.

The effective Hamiltonian would still be extremely complicated if we retained the full band structure for $\varepsilon_n(\mathbf{K})$. However, we have already assumed that the wave function is drawn from only a small region of $\mathbf{K}$ -space, and we can simplify $\varepsilon_n(\mathbf{K})$ to be consistent with this. For example, the bottom of the conduction band in GaAs is approximately

$$\varepsilon_n(\mathbf{K}) \approx E_c + \frac{\hbar^2 K^2}{2m_0 m_e}, \quad (3.12)$$

and the replacement $\mathbf{K} \rightarrow -i \nabla$ gives

$$\varepsilon_n(-i \nabla) \approx E_c - \frac{\hbar^2}{2m_0 m_e} \nabla^2. \quad (3.13)$$

Substituting this into the effective Schrödinger equation (3.11), and moving E_c onto the right-hand side, gives

$$\left[-\frac{\hbar^2}{2m_0 m_e} \nabla^2 + V_{\text{imp}}(\mathbf{R}) \right] \chi(\mathbf{R}) = (E - E_c) \chi(\mathbf{R}). \quad (3.14)$$

This is exactly the result that we sought: a Schrödinger equation that resembles that for free electrons, except for the effective mass, with energy measured from the bottom of the conduction band.

In the case of a donor, with an attractive Coulomb potential, the Schrödinger equation becomes

$$\left(-\frac{\hbar^2}{2m_0 m_e} \nabla^2 - \frac{e^2}{4\pi \epsilon_0 \epsilon_b R} \right) \chi(\mathbf{R}) = (E - E_c) \chi(\mathbf{R}). \quad (3.15)$$

This is exactly like a hydrogen atom except for the effective mass m_e , the offset in energy from the conduction band E_c , and the dielectric constant of the semiconductor ϵ_b (it seems natural that this should appear although we have not proved



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